

# Practical Number 9

## Docker Volumes Demonstration

### Part 1: Life Without Docker Volumes

#### Step 1.1 — Run a container without a volume

```
docker run -d --name mysql-no-volume -e
MYSQL_ROOT_PASSWORD=root123 mysql:8.0
```

```
Microsoft Windows [Version 10.0.26100.4349]
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C:\Users\SBUP>docker run -d --name mysql-no-volume -e MYSQL_ROOT_PASSWORD=root123 mysql:8.0
Unable to find image 'mysql:8.0' locally
8.0: Pulling from library/mysql
edf85873f64e: Pull complete
297d04cfe470: Pull complete
e3e5d1ac74c1: Pull complete
6ef6c7b50a93: Pull complete
49ec2dab01d9: Pull complete
4c8a3e0d4e4b: Pull complete
a62160a5eda1: Pull complete
ab24264a27e9: Pull complete
0d74d296605b: Pull complete
96d30d9fbee8: Pull complete
7534d1db9f8d: Pull complete
Digest: sha256:7dcddc01f13bab2f15cde676d44d01f61fc9f99fe7785e86196dfc07d358ae2b
Status: Downloaded newer image for mysql:8.0
eb72a8eccf3541d05cf42c14ac1c084f5d66e6ec1014db48e2631a19f056ce9f1

C:\Users\SBUP>docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                    NAMES
eb72a8eccf35   mysql:8.0   "docker-entrypoint.s..." 3 minutes ago  Up 3 minutes  3306/tcp, 33060/tcp      mysql-no-volume

C:\Users\SBUP>
```

#### Step 1.2 — Connect and create some data

```
docker exec -it mysql-no-volume mysql -uroot -proot123
```

### Inside the MySQL shell:

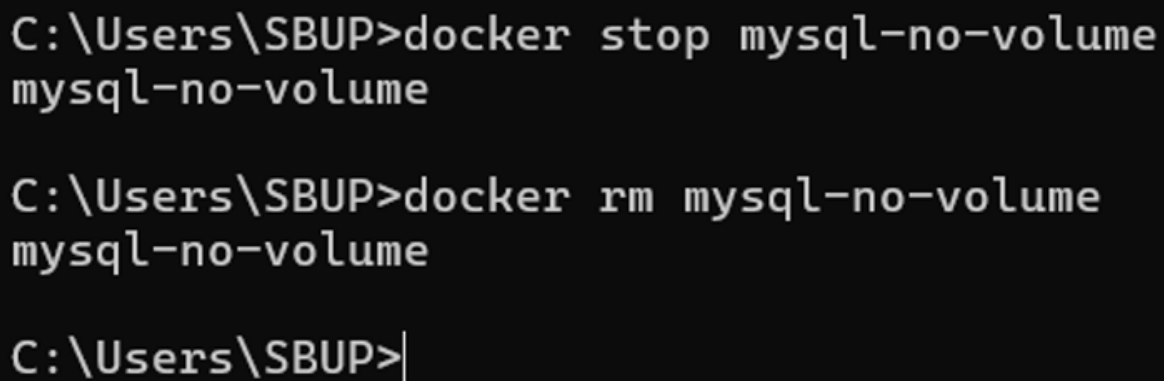
```
CREATE DATABASE testdb;
USE testdb;
CREATE TABLE users (id INT, name VARCHAR(50));
INSERT INTO users VALUES (1, 'Alice');
SELECT * FROM users;
```

### Exit the shell:

```
EXIT;
```

## Step 1.3 — Stop and remove the container

```
docker stop mysql-no-volume
docker rm mysql-no-volume
```



```
C:\Users\SBUP>docker stop mysql-no-volume
mysql-no-volume

C:\Users\SBUP>docker rm mysql-no-volume
mysql-no-volume

C:\Users\SBUP>|
```

## Step 1.4 — Recreate the container and check for data

```
docker run -d --name mysql-no-volume -e
MYSQL_ROOT_PASSWORD=root123 mysql:8.0
docker exec -it mysql-no-volume mysql -uroot -proot123 -e "SHOW
DATABASES;"
```

```
C:\Users\SBUP>docker exec -it mysql-no-volume mysql -uroot -proot123 -e "SHOW DATABASES;"
mysql: [Warning] Using a password on the command line interface can be insecure.
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
```

## Part 2: Life With Docker Volumes

Now let's fix it using a named volume.

### Step 2.1 — Create a named volume

```
docker volume create mysql-data
```

```
C:\Users\SBUP>docker volume create mysql-data
mysql-data
```

Verify it exists:

```
docker volume ls
```

```
C:\Users\SBUP>docker volume create mysql-data
mysql-data

C:\Users\SBUP>docker volume ls
DRIVER      VOLUME NAME
local      1be2f764155d96d9031daee8e2c00379308841e6811ddf9d8b192937e13a68ac
local      7aca080b2e88c199e52f2983b36ab5aa158b3499c39fcbe78b71356958659fc0
local      8b184811f220489aaf3735f672d28c6356cf617b2ff332b466acd1c51725e48f
local      82df3941dceceb0a074fa89e029949a28aa489d96a22b32739581f51fffe436b
local      83b8b61e791143cd8f3ce8dfdc68050447de00991ddd7f28892314f44836e401
local      989853a9a98f83a8b591b1b71fff90fddd794caa44b1ff7f43a7ba4a1d8443fa
local      counter_data
local      db73d28ab5017e033a9db02f5a86ffeac252ff84d8e30bf89cf583e06913e3a5
local      ef0733d4758c1047cb2458bebe3d8c4ac94074f2129ce1c722f11ce216a7cf60
local      jenkins_home
local      mydata
local      mysql-data

C:\Users\SBUP>
```

## Step 2.2 — Run a container using that volume

```
docker run -d --name mysql-with-volume \  
  -e MYSQL_ROOT_PASSWORD=root123 \  
  -v mysql-data:/var/lib/mysql \  
  mysql:8.0
```

```
C:\Users\SBUP>docker run -d --name mysql-with-volume -e MYSQL_ROOT_PASSWORD=root123 -v mysql-data:/var/lib/mysql mysql:8.0  
ce3aa2d53c17d89dccc9210960690e8ad89e98d4d30362a1a9b02262753bf12b  
C:\Users\SBUP>
```

## Step 2.3 — Create data again

```
docker exec -it mysql-with-volume mysql -uroot -proot123
```

```
C:\Users\SBUP>docker exec -it mysql-with-volume mysql -uroot -proot123  
mysql: [Warning] Using a password on the command line interface can be insecure.  
Welcome to the MySQL monitor.  Commands end with ; or \g.  
Your MySQL connection id is 8  
Server version: 8.0.46 MySQL Community Server - GPL  
  
Copyright (c) 2000, 2026, Oracle and/or its affiliates.  
  
Oracle is a registered trademark of Oracle Corporation and/or its  
affiliates. Other names may be trademarks of their respective  
owners.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
mysql> |
```

```
CREATE DATABASE testdb;  
USE testdb;  
CREATE TABLE users (id INT, name VARCHAR(50));  
INSERT INTO users VALUES (1, 'Alice');  
SELECT * FROM users;  
EXIT;
```

```
mysql> CREATE DATABASE testdb;
Query OK, 1 row affected (0.06 sec)

mysql> USE testdb;
Database changed
mysql> CREATE TABLE users (id INT, name VARCHAR(50));
Query OK, 0 rows affected (0.18 sec)

mysql> INSERT INTO users VALUES (1, 'Alice');
Query OK, 1 row affected (0.05 sec)

mysql> SELECT * FROM users;
+-----+-----+
| id    | name  |
+-----+-----+
| 1     | Alice |
+-----+-----+
1 row in set (0.00 sec)

mysql> EXIT;
Bye
```

## Step 2.4 — Destroy the container (but not the volume)

```
docker stop mysql-with-volume
docker rm mysql-with-volume
docker volume ls
```

```
C:\Users\SBUP>docker rm mysql-with-volume
mysql-with-volume

C:\Users\SBUP>docker volume ls
DRIVER      VOLUME NAME
local      1be2f764155d96d9031daee8e2c00379308841e6811ddf9d8b192937e13a68ac
local      7aca080b2e88c199e52f2983b36ab5aa158b3499c39fcbe78b71356958659fc0
local      8b184811f220489aaf3735f672d28c6356cf617b2ff332b466acd1c51725e48f
local      82df3941dceceb0a074fa89e029949a28aa489d96a22b32739581f51fffe436b
local      83b8b61e791143cd8f3ce8dfdc68050447de00991ddd7f28892314f44836e401
local      989853a9a98f83a8b591b1b71fff90fddd794caa44b1ff7f43a7ba4a1d8443fa
local      counter_data
local      db73d28ab5017e033a9db02f5a86ffeac252ff84d8e30bf89cf583e06913e3a5
local      ef0733d4758c1047cb2458bebe3d8c4ac94074f2129ce1c722f11ce216a7cf60
local      jenkins_home
local      mydata
local      mysql-data
```

## Step 2.5 — Recreate the container with the same volume

```
docker run -d --name mysql-with-volume \  
  -e MYSQL_ROOT_PASSWORD=root123 \  
  -v mysql-data:/var/lib/mysql \  
  mysql:8.0
```

```
docker exec -it mysql-with-volume mysql -uroot -proot123 -e  
"SHOW DATABASES; SELECT * FROM testdb.users;"
```

```
C:\Users\SBUP>docker exec -it mysql-with-volume mysql -uroot -proot123 -e "SHOW DATABASES; SELECT * FROM testdb.users;"  
mysql: [Warning] Using a password on the command line interface can be insecure.  
+-----+  
| Database |  
+-----+  
| information_schema |  
| mysql |  
| performance_schema |  
| sys |  
| testdb |  
+-----+  
+-----+-----+  
| id | name |  
+-----+-----+  
| 1 | Alice |  
+-----+-----+  
C:\Users\SBUP>
```